Hassan Mohamed

Systems Builder — Backend · Business Platforms · ERP · SaaS · Desktop Software · Open Source

EMAIL hasanmo3080vr@gmail.com
PHONE +20 110 439 1245
GITHUB github.com/HMH-3080
LINKEDIN linkedin.com/in/hasan-mohamed-81a187289
LOCATION Egypt · open to remote work

PROFESSIONAL SUMMARY

Independent software engineer who designs, builds, deploys and maintains complete systems — architecture, backend, databases, interfaces **and** the production servers they run on. Since 2022 I have delivered **15 production systems** for retail, pharmacy, industrial-supplies and e-commerce operations and still maintain them, many engineered **offline-first** so multi-branch businesses keep trading through internet outages and reconcile cleanly on reconnect. I also **create and maintain my own open-source software** — most notably the Altair web framework for Crystal and the Marsad VPS fleet platform — and contribute to projects owned by others, with **eight merged pull requests** including the Elixir compiler and Ktor.

TECHNICAL SKILLS

Programming languages. Kotlin · Java · TypeScript · JavaScript · Elixir · Crystal · Rust · Python · SQL

Frameworks & application development. Phoenix/LiveView · Erlang/Elixir OTP · Spring Boot · Ktor · Vert.x · Express.js · http4k · HTMX · Kotlin Multiplatform · Altair (Crystal)

Databases & data management. PostgreSQL · MySQL/MariaDB · MongoDB · SQLite · Turso (MVCC concurrent writes)

Infrastructure & operations. Linux server administration · Docker · nginx · systemd · networking · k6 load testing

Architecture & engineering practices. Offline-first synchronization & conflict resolution · multi-tenant isolation · RBAC & audit trails · compile-time-validated design · CI governance gates · benchmarking & profiling

SELECTED PROJECTS & ENGINEERING EXPERIENCE

BUILT, DEPLOYED & MAINTAINED IN PRODUCTION

Business Platforms & Production Systems

INDEPENDENT ENGAGEMENTS

El Fath Pharmacies — multi-branch pharmacy POS & loyalty platform — 2024 · LIVE

- Built an **offline-first multi-branch POS**: batch/lot inventory with expiry monitoring, automated replenishment alerts, cross-branch stock synchronization and full product-movement audit history.
- Implemented **conflict-free cloud sync over local-first persistence**, so branches continue selling through internet outages and reconcile on reconnect.
- Designed **granular role-based access control** with multi-level authorization, activity logging and operational audit trails across branches.
- Delivered POS with automated invoicing, financial reporting with margin monitoring, and a loyalty CRM.

GAL Factory Supplies — industrial supplies enterprise platform — 2023 → present · LIVE

- Delivered a **cloud enterprise platform** for remote operations: sales, inventory, role-based access and secure workflow monitoring across distributed teams (galimport.shop).
- Deployed and hardened the production infrastructure; **continuously own the servers, security and maintenance contract** since launch.

El Tawfikia Auto Parts Markets — automotive spare-parts trading system — 2022 → present · MAINTAINED

- Centralized sales, warehouse inventory, employee operations and workflow tracking on a **secure internal network** with real-time monitoring.
- Continue to provide **hardware maintenance and server administration** for the delivered system to this day.

Pioneer General Supplies & Contracting — business management platform — 2026

- Integrated platform covering sales & invoicing, customer and supplier management, employee roles and cross-department workflow control.

SaaS & Commercial Products

OWN PRODUCTS

Rocket System 2 — distributed ERP + mobile companion — 2026 · LIVE

- **Multi-location warehouse ERP**, offline-first with cloud sync and conflict-free reconciliation; companion mobile app for remote monitoring.
- Deployable on **local servers or the cloud** per customer infrastructure needs.

Egyfy — multi-tenant SaaS e-commerce builder — 2025 → present · BETA

- Architected **tenant isolation**, a SaaS subscription model and a store builder with production-grade scalability.
- **3 tenants live in production**; customer feedback drives the roadmap toward commercial launch.

Rocket System 1 — retail POS & inventory — 2025

- Lightweight **desktop retail solution** for small shops and warehouses — inventory, automated invoices and reporting; designed for teams with limited technical experience.

OPEN-SOURCE PROJECTS — CREATED & LED

MY OWN REPOSITORIES

Altair — web framework for Crystal CREATOR · SOLE BUILDER — MIT · PHASES 0–7

- Designed and built the framework end to end: a **compile-time routing DSL** with typed action references, typed ECR views/layouts/partials and per-request controllers with typed params — invalid programs fail **at compile time, not in production**.
- Implemented a **dual-adapter ORM** (SQLite3 / PostgreSQL) validated through shared contract tests, plus migrations, validations, associations and callbacks.
- Shipped the CLI toolchain (new / generate / server / routes / db), automatic development hot-reload watcher, sessions, CSRF protection, JWT authentication and background jobs.
- Verified by a **~992-spec automated suite** across the stack; performance measured with public **k6 benchmarks** against Express.js and Fiber, run through a dedicated benchmark collector.

Repository · Documentation

Marsad — self-hosted VPS fleet management SOLE BUILDER — 2026 → present · CC BY-NC 4.0

- Browser-based **LiveView mini-OS desktop** in Elixir/Phoenix: one **supervised GenServer with persistent SSH** per managed server (Registry + DynamicSupervisor).
- xterm.js terminal over persistent PTYs (vim, pipes, job control), SFTP explorer with its own query language, inline remote editing and an offline mode.
- Docker / systemd / nginx control panels with SSL-expiry monitoring and per-core live metrics; **AES-256-GCM credential vault** (OTP `:crypto`) with TOFU host-key records.
- Hardened remote execution: **allow-list regex validation** on every value interpolated into shell commands.

Repository

EgyFy Parser — local OCR workspace — Apache-2.0

- Converts PDFs and images into Markdown, JSON and structured tables with per-page confidence review and a clean-output mode; runs as a single-process Python workspace — **documents never leave the machine**.

Desktop, Automation & Utility Projects

PERSONAL OPEN SOURCE

KPM · EG-ID Checker · Pomodoro Timer — Kotlin · Kotlin · RUST

- **KPM** — Maven-backed package access behind an npm-like CLI for the JVM (Kotlin).
- **EG-ID Checker** — Egyptian national-ID validation & demographics library (Kotlin).
- **Pomodoro Timer** — native Windows tray application with toasts and focus screen-blocking (Rust / egui).

OPEN-SOURCE CONTRIBUTIONS — EXTERNAL PROJECTS

MAINTAINED BY OTHERS · MERGED & ONGOING

ASN — Open Social Network (DevHustlers) LEADING CONTRIBUTOR

AGPL-3.0

- **Leading contributor by commit count** to an Elixir-first service monorepo spanning communities, messaging and real-time features: event broker, typed contracts, PostgreSQL persistence.
- Strengthened project governance with **changelog-enforcing CI gates**, contract-drift checks and CODEOWNERS review.

Upstream fixes merged

REPRODUCE → PATCH → TEST → PR → MERGED

- `elixir-lang/elixir #15770` — fixed a type-checker crash on non-returning `for … into:` clauses (Elixir compiler).
- `ktorio/ktor #5633` — corrected CORS default-methods filtering (Kotlin Ktor framework).
- `LiveCodes/browser-elixir #1` — delivered the build toolchain for single-threaded AtomVM targets.
- `loco-rs/loco #1785` — contributed an Arabic translation of the project README (Rust Loco framework).
- `Racore #2` — fixed an LRU-eviction memory leak in `inMemoryIndexes`.
- `404-coffee-backend #1` — added core and user-management test coverage.

TECHNICAL WRITING

- **Nine published engineering articles**: BEAM/Elixir internals and the memory model, distributed-systems design, Kotlin Multiplatform in production, and SQLite MVCC concurrent writes.

LANGUAGES

- **Arabic** — native.
- **English** — professional working proficiency.